Sunny Chang Lu

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Education

Shanghai Starriver Bilingual School Shanghai, China
Class of 2027 | *GPA: 4.770 (weighted)* | *TOEFL: 116*
AP Research (2026–27); 11 of 12 AP exams scored 5.
Audited CIS 160 and CIS 121 at University of Pennsylvania; COS 126 at Princeton University.
In CIS 121, scored 99.0/121.0 on the final (mean 59.14, SD 20.8)
Summer Programs: Ross '26 — six-week, selective proof-based mathematics program
ESPR '25 — European Summer Programme on Rationality (game theory, cryptography, mathematical logic)
Yau Math Camp '25 | PACT Year-Round '21, '22 | Yingcai Talent Program '25 (SJTU CS; advisor Yu Yong)

Research

Kempe's Universality Theorem on Restricted Topologies 2025–Present
Proved that linkages remain universal when restricted to crossing-free, bipartite graphs of maximum degree three; built interactive Desmos constructions of gadgets: Gallery 1, Gallery 2.

HeXO is PSPACE-Hard 2025
Proved $\mathrm{Connect}(n, 6, 2, 1)$ *on a hexagonal grid PSPACE-complete by reduction from Quantified 3SAT.*

Folding Polyominoes into Uniform-Thickness Rectangles 2025–2026
Provided results for solvability as Origami Puzzles for polyominos up to hexominos.

Tiling Squares with Polyominoes 2026
Conclusively characterized which squares each polyomino, up to pentominos, can tile.

PowerPoint Vexillology *(Arglin's Cenote)* 2025–Present
Contributed to a 2,000+ member research effort constructing every flag with PowerPoint tools; supplied the SmartArt cycle construction necessary for flags of Croatia and Kyrgyzstan; presented the work in a talk on the Math Club channel.

Awards & Qualifications

Mathematics

Silver Medal, Distinction and Best in School, British Mathematical Olympiad 2026
STEP II: 115/120 (S); STEP III: 102/120 (S), Sixth Term Examination Papers (STEP) 2026
15th Overall Individual and 11th Theme Round, Harvard–MIT Mathematics Tournament 2023
also 18th Combinatorics (2025) and 23rd Geometry (2026)
AIME: 12/15 (2025, 2026); AMC 12: 138/150 (2026); AMC 10: perfect score, top 1% (2023) 2023–2026
1st Individual and 1st Team, Stanford Math Tournament 2025
1st Individual and 1st Team (2024); 2nd Individual, High Distinction (2025), Yale MMATHS 2024–2025
Team 3rd Overall; Individual 8th, Combinatorics 3rd, Princeton University Mathematics Competition 2026
Global Top-10 Team and National 3rd, High Distinction, Math Tournament at Berkeley 2024
National Team Gold, National Individual Top 100; Power Round I Silver, ARML Local 2023
1st Team Overall and 2nd Individual, Duke Math Meet 2025
also 1st Team Overall, 1st Team Challenge and Relay, 3rd Individual in regional qualifiers
Honorable Mention, International Mathematical Modeling Challenge 2026

Informatics & Artificial Intelligence

Silver Medal, Scientific Round, International Olympiad in Artificial Intelligence 2024
Gold Division, USA Computing Olympiad 2023
First Prize, National Olympiad in Informatics in Provinces (NOIP) 2025
CSP-J and CSP-S: First Prize, both rounds (CCF informatics olympiad) 2023
Winner, 2nd International Obfuscated Python Code Contest 2024
All-Star, Global Bronze, ACSL Programming Contest 2023

Physics & Chemistry

Global Gold; Global Top Gold (top 2%); Round 2 Silver, British Physics Olympiad 2024–2026
Global Gold (49/50), Senior Physics Challenge 2025

National Gold (2025); National Silver (2024), Physics Bowl Division 1 2024–2025
Global Gold, Cambridge Chemistry Challenge (C3L6) 2024

Linguistics, Humanities & Design

High Distinction (top 10%), International Linguistics Olympiad Regional Round 2026
John Locke Institute Essay Competition 2023–2024
Commendations in Politics and Philosophy (2023); Merit in Politics (2024)
Finalist, Best Pen & Paper Puzzle, Thinky Awards 2025

Activities & Leadership

OEIS Contributor 2025–Present
Proposed sequence A398486; contributed corrections and comments to existing entries
Presenter of 2swap's Math Club 2025–Present
Seminars on tilings, foldings, and PowerPoint constructibility; related videos have 800,000+ total views on YouTube
Open-Source Developer 2023–Present
Origami Simulator, HeXO Renderer, and miscellaneous projects
Puzzle Designer 2023–Present
Co-designed Origami Puzzles, Thinky Awards finalist | co-designed ICNP | contributor to PUZZDLE
playtester for Herd, Workworkwork, and Mondrian Restoration
Speaker, TEDxXuhui 2023
Tentative speaker for 2026 TEDxSSBS
Associate Editor and Writer, nondefault zine 2026–Present
Participant, LIVE Communication Training 2025–Present
Writer, The Curious Incident of the Fork in the Night-Time 2024
fanfiction of Mark Haddon's original The Curious Incident of the Dog in the Night-Time

Work Experience

Worksheet Author, starmathcat's worksheets *(YouTube mathematics channel)* 2026–Present
Mathematics Tutor, Cambridge STEP Exam Preparation *(five sessions)* May 2026
Freelance Translator, *Checkmating Combinations in Chess* into Chinese Aug–Dec 2024

Skills

Programming: proficient in Python, Desmos, and LaTeX | open source work with Kotlin | basic C++ and Java knowledge
Languages: fluent in English, Chinese (Mandarin), and toki pona